

School Bus Tracking Software

Project Flow Document – Version 1

Core Flow: Route → Stops → Student → Parent → Daily Driver/Bus Assignment → Live GPS → ETA → Notifications

1. Driver Flow

- Driver is not permanently assigned to a particular bus or route.
- A driver may drive Route A today and Route B tomorrow.
- Driver logs in using Mobile Number + OTP / Login.
- After login, the driver sees the route/trip assigned for that day.
- Driver confirms the assigned trip and starts the trip.
- The assigned route contains predefined stops.
- When the trip starts, live GPS tracking begins and parent/web notifications are triggered.
- The system automatically updates stop arrival/crossing status.
- Driver ends the trip after completion.

Important Rule

Driver + Bus + Route + Date/Trip = Daily Assignment. There should be no mandatory permanent Driver → Bus → Route mapping.

2. Route Management

- Admin/School Admin creates a route.
- Each route contains a sequence of predefined stops.
- Route fields: Route Name, Route Number/Code, Starting Point, School and Status.
- Stop fields: Stop Name, Latitude, Longitude, Sequence, Pickup Time, Drop Time and Geofence Radius.

Example

School → Gandhipuram → Peelamedu → Hope College → Singanallur → School

3. Driver Daily Assignment

- Transport Manager/Admin assigns the driver, bus, route and trip for a specific date.

Example

07-Oct-2026: Driver Kumar + Bus TN-38-AB-1234 + Route RS-01 + Morning Pickup.

- The next day the same driver may be assigned to a different bus or route.
- Morning Pickup and Evening Drop should be treated as separate trip types.

4. Student Registration & Route Assignment

- Register student with Student Name, Student ID/Admission Number, Class, Section and Parent Details.
- Admin selects the route for the student.
- Only the stops belonging to the selected route should be displayed.

- Admin selects the student's pickup/drop stop.

Important Rule

Student is assigned to a Route and a Stop. Parent does not select an arbitrary route.

5. Parent Login & Multiple Children

- Parent logs in using Mobile Number + OTP.
- After login, all linked children are displayed.
- If a family has multiple children, provide a **Switch Child / Switch Account** option.
- When the child is switched, the route, stop, bus, ETA, trip status and notifications must update for that child.

6. Parent Live Bus Tracking

- Parent app displays the bus on a live map.
- Show current bus location, bus status, speed (optional), distance and ETA.
- Live location should update without requiring the parent to manually refresh the screen.

Example

Bus: TN-38-AB-1234 | **Status:** Moving | **ETA:** 8 minutes | **Distance:** 2.4 km

7. Morning Trip Status & Notifications

Bus Started

"Bus Started from School"

Approaching

"Bus is approaching your stop. Estimated arrival: 8 minutes."

Reached

"Bus reached your stop."

Crossed

"Bus has crossed your stop."

- Reached/Crossed/Upcoming status should be determined automatically using GPS + geofence logic.

8. Return Trip / Evening Drop

- When the bus leaves school: "Bus Started from School."
- While approaching the parent's stop: "Bus is approaching your stop. ETA: 6 minutes."
- When it reaches the stop: "Bus reached your stop."
- If the bus has passed the stop: "Bus has crossed your stop."
- Morning Pickup and Evening Drop should be clearly distinguished in the parent app.

9. Delay Management

- System should detect or allow admin to record delays.
- Examples: 10-minute delay, 12-minute delay or 15-minute delay.
- Possible reasons: Traffic, Bus Breakdown, Road Block, Weather, Driver Issue or Other.

Example Notification

"Bus Delayed by 15 Minutes. Please expect a delay in reaching your pickup location."

10. Bus Breakdown / Emergency

- Admin/Transport Manager can send an emergency notification to affected parents.

Example

"Important Bus Update: Your school bus has experienced a breakdown. Please wait for further instructions from the school."

Pickup Change

"Parents are requested to pick up their children from school."

11. School Notification / Announcement System

- Admin can create a notification with Title, Message, optional Attachment and Target.
- Target must support two levels: Entire School and Specific Route.

Entire School

Notification is sent to all students/parents across all routes.

Specific Route

Notification is sent only to parents/students assigned to the selected route.

12. Notification Types

Automatic Notifications

Bus Started, Bus Approaching, Bus Reached Stop, Bus Crossed Stop, Bus Delayed, School Reached and Trip Completed.

Manual Notifications

School Announcement, Route Announcement, Bus Breakdown, Traffic Delay, Pickup Change, Emergency Message, Holiday and Other Important Information.

13. Overall Project Flow

Step	Flow
1	ADMIN → Create Routes
2	Route → Add Stops
3	ADMIN → Register Students
4	Student → Assign Route → Assign Stop
5	ADMIN → Register Drivers & Buses
6	Daily Assignment → Driver + Bus + Route + Trip
7	DRIVER APP → Login → Confirm Trip → Start Trip
8	Driver GPS → Go Backend → Live Location
9	Live Location → Parent App + Web Admin
10	GPS + Geofence → ETA / Upcoming / Reached / Crossed
11	Trip Events → Automatic Notifications
12	Admin → School-wide / Route-specific Notifications
13	Driver → End Trip

14. Core Business Rules

- Driver assignment is daily; driver is not permanently tied to a route or bus.
- Every route contains ordered stops.
- A student belongs to a route and a specific stop.
- A parent can have multiple children and can switch between children.
- Live bus location is shared with the correct parents based on the child's assigned route/stop.
- Reached and Crossed status must be GPS/geofence driven.
- Notifications can target the entire school or a specific route.
- Morning Pickup and Evening Drop are separate trip contexts.
- ETA should be calculated from the current bus position to the student's assigned stop.